Run report: inherited-F4 eager rewrite, `f6_ic_probe`, run-20261011T1657Z

Observations only. Interpretation is in the study report (`../../REPORT.pdf`).

What ran

- **Binary:** `examples/f6_ic_probe` built in release from commit `6bdf93f4b` (branch `groebner-f4f5f6-audit-20261010`; its `src/cryptanalysis/inherited_f4.rs` is byte-identical to the PR branch `inherited-f4-eager-rewrite`). One binary for both arms; the arm is the environment variable `KIC_F4_EAGER` (`0` = lazy rows, unset/`1` = eager rewrite).
- **Commands** (one per file in `raw/`):
  - `KIC_F4_EAGER=$e f6_ic_probe --a 0 --n 23 --dim 8 --targets 4 --engines f4,f6p --reps 3`
  - `KIC_F4_EAGER=$e f6_ic_probe --a 0 --n 31 --dim 9 --targets 1 --engines f4,f6p --reps 3`
- **Order:** three rounds; in each round lazy then eager, n = 23 then n = 31 (interleaved).
- **Host and isolation:** Claude Code cloud container, 4 vCPU Intel Xeon @ 2.10 GHz (AVX-512, BMI2), shared host, no core pinning, no frequency control. Isolation level: none (L0). Single-threaded probe.
- **Workload:** Koblitz curve K_0 over $\mathbb{F}_{2^n}$, standard-subspace factor base, $m = 3$, targets $[k]G$, $k = 1\ldots$, node budget 20 000. Arms within the probe: `f4` (inherited F4) and `f6p` (F6-IC with the shared pair-sum index).
- **Requested / completed:** 12 probe invocations requested, 12 completed, no failures.
- **Unit:** `total_ms` is the probe's own wall time per arm: for each target the best of 3 repetitions, summed over targets (milliseconds, measured).

Recorded values

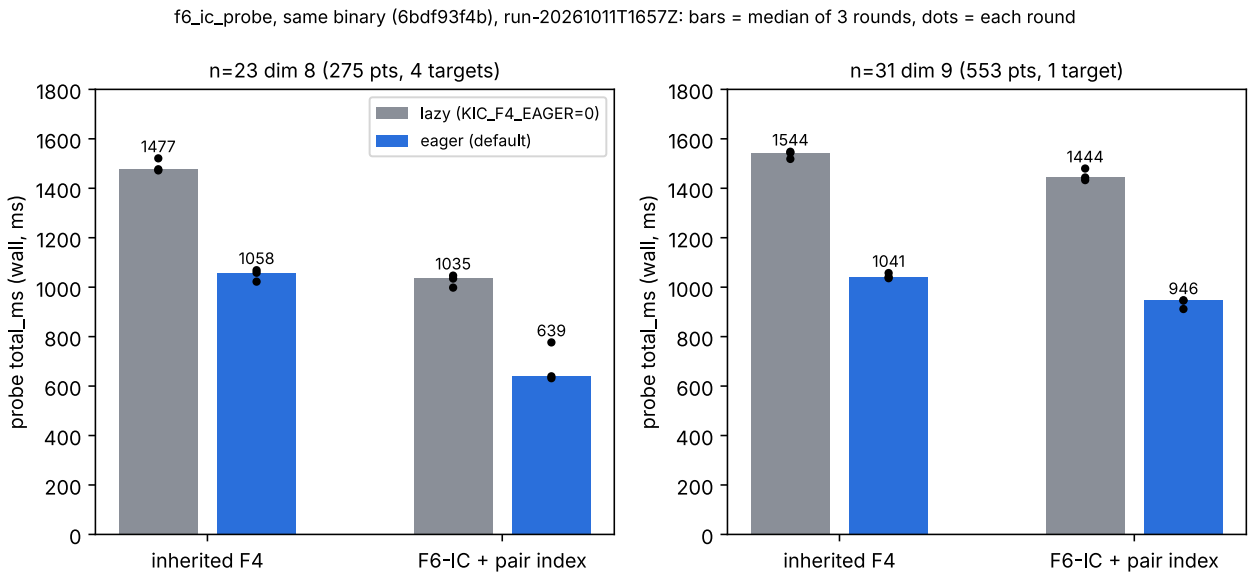


Figure 1: Probe `total_ms` per arm and mode, run-20261011T1657Z, 3 rounds each (dots); bars are the median of the 3 recorded values.

cell	arm	mode	round 1	round 2	round 3
n23 dim 8	f4	lazy	1472.2	1521.1	1477.0
n23 dim 8	f4	eager	1058.4	1068.6	1022.3
n23 dim 8	f6p	lazy	998.2	1034.8	1046.8
n23 dim 8	f6p	eager	631.8	776.6	639.5
n31 dim 9	f4	lazy	1543.5	1547.8	1518.7
n31 dim 9	f4	eager	1057.2	1036.3	1041.3
n31 dim 9	f6p	lazy	1433.0	1443.6	1479.9
n31 dim 9	f6p	eager	946.8	946.1	911.6

All values measured, ms. Derived: medians of the three rounds (shown above the bars) are n23 f4 1477.0 / 1058.4, n23 f6p 1034.8 / 639.5, n31 f4 1543.5 / 1041.3, n31 f6p 1443.6 / 946.1 (lazy / eager).

Counters (identical in every round)

cell	arm	reductions	splits	word_xors lazy	word_xors eager
n23 dim 8	f4	2796	1192	158 512 775	179 177 701
n23 dim 8	f6p	1045	636	124 949 801	144 396 510
n31 dim 9	f4	1410	602	281 583 026	310 429 172
n31 dim 9	f6p	528	323	278 237 634	305 397 167

Reductions and splits are equal between modes; every target's verdict (`hit`, `exhausted`) is equal across modes and rounds (10 arm–target pairs, all `hit: false`, `exhausted: false`). `word_xors` differs by mode because the eager arm charges the rewrite of every kept row.

Records

`raw/n23_dim8_eager{0,1}.round{1,2,3}.jsonl`, `raw/n31_dim9_eager{0,1}.round{1,2,3}.jsonl` (probe stdout, one JSON line per arm and target plus totals and the solver profile). Graph source: `plot_totals.py` (reads `raw/` only).

Missing

No repeated sessions on an isolated host; no interval beyond the three rounds shown. The n23 `f6p` eager round 2 value (776.6 ms) is recorded as it came; no rerun.